

● HARD

● INFORMATION AND IDEAS

● ANSWER KEY

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# Information and Ideas

# 07

10 answers · Rationale-rich · Teach from doc



Q1

C

- C. “Stevens’s ambassadorial sandpaintings are best understood not as self-contained objects but as reminders of the public creations of the sandpaintings, during which Stevens conducted appropriate *ikaah* rituals and encouraged viewers to closely track his movements and subtle shifts in the sand throughout the process.”

Choice C is the best answer because it most directly supports the text’s claim that Stevens strove to convey the ephemeral and dynamic characteristics of *ikaah* as a cultural practice. The text challenges the characterization of Stevens’s preserved sandpaintings as mere “static objects” that cannot “authentically represent” a practice defined in part by the absence of an end product; it claims instead that this “overly object-focused” interpretation overlooks how Stevens preserved the essential characteristics of traditional *ikaah* practice. This quotation supports the claim by explaining that Stevens’s sandpaintings should be understood “not as self-contained objects but as reminders of the public creations of the sandpaintings,” during which Stevens “conducted appropriate *ikaah* rituals and encouraged viewers to closely track his movements and subtle shifts in the sand throughout the process.” The fact that Stevens created the sandpaintings in a public setting and invited viewers to attentively engage with this process supports the claim that he sought to portray *ikaah* as a dynamic practice rather than as a static product. Further, the inherently impermanent nature of such a performance evokes the ephemerality that characterizes *ikaah*.

Choice A is incorrect because it contradicts rather than supports the text’s claim that Stevens strove to convey the ephemeral and dynamic characteristics of *ikaah* as a cultural practice. This quotation argues that Stevens’s preserved sandpaintings “should not be confused with authentic *ikaah*” and emphasizes that authentic *ikaah* “cannot be extricated from a practice that is intentionally and necessarily transitory.” This view aligns with the “overly object-focused” perspective that the text argues against, rather than supporting the claim that Stevens successfully conveyed *ikaah*’s ephemeral and dynamic characteristics. Choice B is incorrect because it focuses on reconciling the tension between ephemeral and permanent forms rather than supporting the claim that Stevens actively conveyed *ikaah*’s ephemeral and dynamic characteristics. While this quotation acknowledges an apparent contradiction, it suggests that Stevens was accommodating US preferences for permanent artworks; it doesn’t demonstrate how he communicated the ephemeral and dynamic nature of traditional *ikaah* practice. Choice D is incorrect. Although the quotation mentions Stevens’s “temporary and performative role” as a gift-giver, it doesn’t address how Stevens conveyed the ephemeral and dynamic characteristics of *ikaah* itself during his ambassadorial activities. The quotation focuses on Stevens’s role in transmitting knowledge and representing cultural dynamism but doesn’t specifically support the claim about his efforts to communicate the essential characteristics of *ikaah* practice. Further,

the quotation incorrectly implies that Stevens shared his knowledge of preservative additives; the text doesn't establish that he revealed how the sandpaintings were preserved.

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Q2

D

- D. rather than undergoing changes resulting in the eventual elimination of a head region in their radial body plan, as previously assumed, sea stars' morphology evolved to completely lack a trunk and consist primarily of a head region.
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Choice D is the best answer because it most logically completes the text's discussion of the morphology (form and structure) of sea stars, a type of echinoderm. The text indicates that echinoderms have radially symmetrical body plans (symmetrical around a central point, usually in the form of a star), whereas most animals have bilaterally symmetrical body plans (symmetrical along an axis running from head to tail through a trunk). According to the text, sea stars are unusual echinoderms because, despite their radial body plan, they descended from known bilateral ancestors. This shift in body plan was thought to be a process of losing the genetic markers associated with the head region. The text explains that by comparing the genes of one sea star species (*P. miniata*) to those of a close relative, the acorn worm, researchers determined that instead, anterior (head) genes are active across the sea star's entire body, posterior (tail) genes are active in limited, peripheral locations of the body, and no trunk-related genes are active. This finding strongly suggests that, rather than becoming "headless" as they evolved from a bilateral ancestor, sea stars developed a body plan consisting almost entirely of a head region with a minimal tail region and no trunk region present.

Choice A is incorrect because the text doesn't identify how any particular region of sea stars' bodies influences the layout of sea stars' radial symmetry. Moreover, the text indicates that the radial symmetry of echinoderms is "usually starlike," not that a starlike layout distinguishes sea stars from other echinoderms. Choice B is incorrect because the text doesn't suggest that the idea that sea stars evolved from an ancestor with bilateral symmetry is incorrect (describing the bilateral origin as "known") and doesn't address any body plans other than those with radial or bilateral symmetry. The text strongly suggests that rather than revealing something about sea stars' origin, Formery et al.'s findings contradict the assumption that the current body plan of sea stars is "headless." Choice C is incorrect because the text suggests that Formery et al. were able to make determinations about *P. miniata* sea stars' body plan based on the comparability of genetic markers between *P. miniata* and *S. kowalevskii* acorn worms. The text indicates only that little or no activity was observed in certain types of genes associated with body development in *P. miniata*, not that those genes turned out to largely differ from body-development genes in *S. kowalevskii*.

Q3

A

- A. They likely make use of different types of evidence than historians of capitalism typically rely on.
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Choice A is the best answer because it presents a statement about Alfagali and other Africanist scholars that is supported by the text. The last sentence of the text explains that historians of capitalism generally rely on a type of data that requires "archival records, which do not exist for much of precolonial Africa." Nevertheless, Alfagali and other Africanist scholars have demonstrated that features of capitalism including "returns-focused investment,...open markets for wage labor, and other features of capitalism" existed in parts of Africa before colonization. Because archival records typically used by historians of capitalism aren't available for precolonial Africa, Alfagali and other Africanist scholars of precolonial capitalism in Africa likely used atypical sources for their studies.

Choice B is incorrect because the text states that archival records containing longitudinal economic data "do not exist for much of precolonial Africa." If these records don't exist for precolonial Africa, then Alfagali and other Africanist scholars couldn't derive data from them by any method. Choice C is incorrect because the text doesn't suggest that Alfagali and other Africanist scholars define capitalism differently than other historians of capitalism do. On the contrary, the text simply asserts that specific features they found, such as "returns-focused investment [and] open markets for wage labor," are "features of capitalism," which implies scholarly agreement that these activities are capitalist in nature. Choice D is incorrect because the text doesn't indicate any effects of capitalism in Africa, positive or negative. Rather, the text explains why precolonial Africa has been omitted from mainstream histories of capitalism and presents work by Alfagali and other Africanist scholars that challenges and corrects the historical record of precolonial capitalism.

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- C. The bite force estimates produced by Bates and Falkingham and by Cost et al. were similar to each other, while the estimates produced by Meers and by Gignac and Erickson each differed substantially from any other estimate.

Choice C is the best answer because it accurately describes data from the table that support Barrett and Rayfield's suggestion about bite force estimates. According to the text, Barrett and Rayfield believe that estimates of dinosaur bite force may be strongly influenced by the methods used to produce them—that is, that different methods may produce significantly different results. The table shows that the studies by Bates and Falkingham and by Cost et al. used the same estimation method (muscular and skeletal modeling) and produced similar bite force estimates (approximately 35,000–57,000 newtons and 35,000–63,000 newtons, respectively). The study by Meers, however, used body-mass scaling and produced a much higher bite force estimate (183,000–235,000 newtons), while the study by Gignac and Erickson used tooth-bone interaction analysis and produced a much lower bite force estimate (8,000–34,000 newtons). The fact that one method produced similar estimates in two different studies and that two different methods used in other studies produced substantially different estimates supports the idea that dinosaur bite force estimates are significantly influenced by the methodology used to produce them.

Choice A is incorrect because it inaccurately describes data from the table. The table does show that the studies by Meers and by Cost et al. used different estimation methods and produced very different ranges of estimated dinosaur bite force, which would support Barrett and Rayfield's suggestion that different methodologies may produce significantly different estimates. However, the table doesn't show that the study by Meers produced the lowest estimated maximum bite force while the study by Cost et al. produced the highest. In fact, the study by Meers estimated a maximum bite force of approximately 235,000 newtons, which is the highest of all the estimated maximums. Choice B is incorrect. Although the data from Gignac and Ericson's study are accurately described, a single set of findings from one study using only one methodology can't show that different methodologies may produce significantly different dinosaur bite force estimates, as Barrett and Rayfield suggest. Choice D is incorrect. Although the table shows that the maximum bite force estimated by Cost et al. was higher than that estimated by Bates and Falkingham, the difference is relatively small; in fact, both teams estimated a minimum bite force of approximately 35,000 newtons and a maximum bite force close to approximately 60,000 newtons. Because these findings demonstrate that a single methodology (muscular and skeletal modeling) produced similar overall results in two studies,

the findings don't support Barrett and Rayfield's suggestion that different methodologies may produce significantly different dinosaur bite force estimates.

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**Q5**

**C**

- C. may have determined that the green plan imposed additional burdens on them that were not sufficiently offset by the potential financial savings.

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Choice C is the best answer because it most logically completes the text's discussion of the influence of decision-making modes on consumers' choices of different electricity plans. The Affect Mode describes choices based on the decision-maker's preferred emotional outcome, the Role Mode describes choices based on the decision-maker's social motivations, and the Calculation Mode describes choices based on minimizing financial costs and maximizing the decision-maker's convenience. Thus, if those using a Calculation Mode disfavor the green plan, it would be due to some financial or convenience burden the green plan imposes that the other doesn't. The text indicates that the green plan had a lower financial cost but was nonetheless rejected by participants using the Calculation Mode. It therefore follows that the green plan likely imposed a convenience burden that outweighed potential financial savings for these participants.

Choice A is incorrect. According to the text, the Affect Mode involves the decision-maker's preferred emotional outcome, the Role Mode involves social motivations, and the Calculation Mode is oriented toward minimizing financial costs and maximizing the decision-maker's convenience. Thus, unlike the Affect or Role Modes, the Calculation Mode would almost certainly involve taking price differences into account. Choice B is incorrect. According to the text, the Affect Mode involves the decision-maker's preferred emotional outcome, the Role Mode involves social motivations, and the Calculation Mode is oriented toward minimizing financial costs and maximizing the decision-maker's convenience. Thus, there is no reason to think that participants using the Calculation Mode at any point took social responsibility into account, let alone think that they discovered a disparity between their expectations and reality. Choice D is incorrect. According to the text, the Affect Mode involves the decision-maker's preferred emotional outcome, the Role Mode involves social motivations, and the Calculation Mode is oriented toward minimizing financial costs and maximizing the decision-maker's convenience. Thus, the text doesn't indicate that cost-effectiveness is a factor for participants using the Affect or Role Modes, and there is no basis to compare their beliefs in the truthfulness of the green-plan description with those of the Calculation Mode users.

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Q6

D

- D. The study by McAllister et al. suggests that when traveling waves intersect in specific ways, the resulting wave may be higher than would be expected based on the properties of traveling waves.
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Choice D is the best answer because it most accurately states the main idea of the text. The text begins by explaining that most studies of ocean wave breaking have focused on traveling waves (those that move horizontally) but that McAllister et al. studied spike waves, which form when traveling waves moving in opposing directions intersect. The text establishes that while traveling waves break when their steepness exceeds a critical threshold (limiting their height), McAllister et al. found that spike waves can exceed these constraints because factors like jet stability and cavity shape can mitigate the effects of steepness. Thus, the main idea of the text is that spike waves resulting from traveling waves intersecting in specific ways can reach heights greater than what would be expected based on the general properties of traveling waves.

Choice A is incorrect because it misrepresents the information in the text; the text indicates that McAllister et al. showed that jet stability and cavity shape (along with steepness) influence breaking in spike waves, not in traveling waves. Further, the information about specific influences is not the main focus of the text. Choice B is incorrect because it misrepresents the information in the text; the text indicates that McAllister et al. found that spike waves can pass the steepness threshold at which traveling waves break because factors like jet stability and cavity shape affect the breaking of spike waves, not because they've simply reached a critical threshold of steepness. Choice C is incorrect. Although the text does explain that spike waves can form when traveling waves intersect, it presents this as a known fact that McAllister et al.'s study was based on, not as something the researchers suggested. Further, the text indicates that spike waves can exceed a threshold that limits traveling wave height but doesn't focus on the idea that spike waves typically are higher than traveling waves.

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Q7

D

- D. For each species, the percentage of juvenile plants growing in patches of vegetation was substantially higher than what would be expected if plants were randomly distributed.

Choice D is the best answer because it provides the most direct support from the table for the claim that the plants growing in close proximity to other plants gained an advantage at an early developmental stage. The table shows the total number of juvenile plants from five species that were found growing on bare ground and in patches of vegetation as well as the percentage of the total number of each species that were growing in patches of vegetation. For each of the five species, more than 50% of the juvenile plants were growing in patches of vegetation. The text notes, however, that a random distribution of plants across the landscape should result in only about 15% of the plants being found in patches of vegetation. In other words, for each of the five species, the percentage of juvenile plants found growing in patches of vegetation was substantially higher than could be explained by chance alone. This finding supports the claim in the text: if plants growing in patches are overrepresented among plants that have survived to the juvenile stage, as the data show they are, then it suggests that it's advantageous for plants at an early stage of development to grow in patches of vegetation.

Choice A is incorrect because the statement that less than 75% of juvenile plants were found growing in patches of vegetation, while true, doesn't clearly support the claim that the plants growing in close proximity to other plants gained an advantage at an early developmental stage. Saying that less than 75% of plants were found in patches doesn't indicate how the percentage growing in patches compares with the percentage that would be expected to grow in patches on the basis of chance alone, which is the information necessary to evaluate whether the claim in the text has support in the table. Put another way, if the percentage of plants found growing in patches was 15% or less, it would be true that less than 75% were found in patches, but the data would in fact weaken the claim in the text, not strengthen it, since the data would show that growing in patches wasn't advantageous. Choice B is incorrect because only 12 plants of this species were found growing in patches, which was the lowest number of any species, not the greatest number. Additionally, even if it were true that this species had the greatest number of plants growing in patches, the finding would be irrelevant to the claim that plants of all five species gained an advantage by growing in close proximity to other plants. Choice C is incorrect because 59.1% of the plants of these species were found growing in patches, which is a far greater percentage, not a lower percentage, than what would be expected if plants were randomly distributed (around 15%). Additionally, if it were true that the percentage of plants growing in patches was lower for these species than what would be

expected from chance alone, that finding would weaken, not strengthen, the claim that growing in patches is advantageous.

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Q8

D

- D. the aptamers successfully bound to *E. coli* O157:H7 and the tests displayed a high degree of specificity in a real-world food.
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Choice D is the best answer because it most logically completes the text's discussion of aptamers' capacity to test for pathogens in food. The text explains that although synthetic aptamer molecules can be used to test for foodborne bacterial pathogens because they bind to target molecules, it hasn't been clear how likely it is that they will indicate a negative result when a target pathogen is absent in real-world foods. The text then indicates that Somvanshi et al. created test paper that changes from pink to purple when aptamers in it bind to a particular strain of *E. coli* bacteria, O157:H7, and tested it with store-bought pear juice (that is, a real-world food); the paper changed to purple when exposed to juice to which the target pathogen *E. coli* O157:H7 had been added, but it remained pink when exposed to juice treated with other *E. coli* strains or other bacteria species. Based on this result, it seems the aptamers in the paper successfully bound to the target (O157:H7) and the tests had a high degree of specificity (providing negative results for samples where the target was absent even though other bacteria were present) when applied to a real-world food.

Choice A is incorrect because nothing in the text suggests that *E. coli* O157:H7 differs from other pathogens in a way that makes it more suitable for aptamer-based testing of any kind and that specificity is likely to be lower when aptamer-based tests target other bacteria; the text simply indicates that Somvanshi et al. used O157:H7 as the target for aptamer-based test paper in their study and suggests that the paper had a high degree of specificity. Choice B is incorrect because the text indicates that the specificity of aptamer-based tests in real-world foods is uncertain for pathogens broadly, not just for *E. coli*, and similarity between strains (of *E. coli* or of other pathogens) isn't mentioned. Moreover, the results presented in the text suggest that aptamers are actually capable of distinguishing between strains of *E. coli*, since Somvanshi et al.'s paper turned purple when exposed to *E. coli* O157:H7 and remained pink when exposed to other strains. Choice C is incorrect because the text suggests that the aptamers in the test papers didn't bind to different strains of *E. coli*. The text explains that the test papers turn purple when the aptamers bind to the targeted pathogen and that in the pear juice tests, the test papers turned purple when exposed to samples with the targeted strain of *E. coli* (O157:H7) but remained pink when exposed to samples with other strains of *E. coli*. In other words, the aptamers bound only to the targeted strain, not to the other strains. Further, specificity would be affected if the aptamers had bound to multiple strains and not just the targeted one; that result would cause the specificity to be low.

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Q9

D

- D. Although the claim of a late ninth- through early eleventh-century composition date for *Beowulf* has some plausibility, advocates for the claim have not compellingly addressed evidence suggesting an earlier date.
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Choice D is the best answer because it most accurately states the main idea of the text. The text states that some scholars have dated the composition of *Beowulf* to the late ninth through early eleventh centuries due to the poem's fit with that period's historical context. The text goes on to say that while it is "not inconceivable that *Beowulf* emerged from such a context"—that is, it is possible that *Beowulf* was composed during the late ninth through eleventh centuries—there is linguistic evidence that the poem was composed earlier, in the seventh or eighth century. According to the text, favoring the historical context over the linguistic evidence requires justification that scholars have not yet supplied. In other words, the text suggests that scholars who favor the later composition date need to explain why the poem's fit with historical context should take precedence over the linguistic evidence, but they have not yet done so. Thus, the main idea of the text is that while there is some plausibility to the later composition date, advocates for the later date have not compellingly addressed evidence suggestive of an earlier date.

Choice A is incorrect because the text says that scholars who date the poem to the late ninth through early eleventh centuries have failed to account for the linguistic evidence that the poem may have been composed earlier, not that the evidence those scholars cite in favor of their view is unreliable or that anyone has cast doubt on that evidence. In other words, the text does not suggest that there are problems with the evidence cited by advocates of the later composition date, only that there is other evidence of an earlier composition date that those advocates need to consider. Choice B is incorrect because nothing in the text suggests that those scholars who date the poem to the late ninth through early eleventh centuries are giving priority to a controversial view of the social conditions at that time. The text makes no reference to any controversy about how scholars interpret that historical period. Instead, the text suggests that scholars who date the poem on the basis of its fit with the historical context of England in the late ninth through early eleventh centuries have failed to account for linguistic evidence that the poem may have been composed earlier. Choice C is incorrect because the text says nothing about how well the poem fits the historical context of England in the seventh and eighth centuries, let alone that it fits that historical context as well as it fits the historical context of the late ninth through early eleventh centuries. Rather, the text says that there is linguistic evidence that the poem may have been composed in the seventh or eighth century.

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## Q10

B

- B. None of the percentages shown for provisioning services are lower than 33%, and the overall percentage shown for provisioning services is 75%.
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Choice B is the best answer. The finding that all the regions reported at least some effects on provisioning services supports the first part of the assertion. And the fact that provisioning services comprise 75% of the reported services overall supports the second part of the assertion.

Choice A is incorrect. This choice doesn't fully support the assertion. It doesn't include the finding that all the regions (not just Central and West) reported at least some effects on provisioning services. Choice C is incorrect. This choice doesn't fully support the assertion. It doesn't demonstrate that provisioning services represent the majority of the reported services overall. Choice D is incorrect. This choice doesn't fully support the assertion. It doesn't demonstrate that provisioning services represent the majority of the reported services overall.

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### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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